

Influence of Functionalized Polypropylene on Polypropylene/Graphene Oxide Nanocomposite Properties

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Graphene oxide (GO) derived from oxidation of natural graphite contains many active groups that can interact with a great variety of polar moieties. In this work, polypropylene (PP)/graphene oxide nanocomposites using polypropylene (PP) grafted with amine-alcohol (PPgDMAE) as compatibilizer were prepared by two different methods. Maleic anhydride grafted PP (PPgMA) was reacted with 2-[2-(dimethylamine)-ethoxy] ethanol (DMAE) in the melt for forming amine-alcohol functionalized polypropylene (PPgDMAE). Nanocomposites were prepared by two methods. In one method, PP/GO nanocomposite was prepared by direct melt mixing in an internal batch mixer using PPgDMAE as compatibilizer. In another method, a previous mixing of PPgDMAE with GO in hot Xylene was done and then, once the solvent was evaporated, it was incorporated into PP by melt-mixing. The microstructure and interface enhancement of the prepared composites were analyzed by Fourier transform infrared spectroscopy (FTIR), Raman, X-ray diffraction (DRX) contact angle, scanning and transmission electron microscopy (STEM), mechanical, thermal, and electrical properties measurements. Fourier transform infrared spectroscopy (FTIR) revealed the interaction between GO and PPgDMAE. The loading of GO con-

ducted to enhance the composite mechanical properties attributed to the strong interfacial interactions between GO and PPgDMAE. A significant improvement in mechanical thermal stability and electrical properties was observed when nanocomposites were prepared by the solution blending method compared with melt mixing method. This work suggests a potential application of GO in preparation of high performance PP composites. POLYM. COMPOS., 00:000–000, 2016. © 2016 Society of Plastics Engineers

INTRODUCTION

Polymer nanocomposites reinforced with nanofillers have gained great interest in the last decades mainly because of the considerable enhancement in mechanical, thermal, and flammability properties using low filler loadings. Loading of nanofillers such as layered silicates montmorillonite (MMT), inorganic nanoparticles (silver, TiO₂, ZnO, etc.), and carbon nanostructures (nanotubes (CNTs), fullerenes, nanofibers (CNF), graphene, etc.) has resulted in the substantial enhancement of mechanical, thermal, electrical, and gas barrier properties at much lower loadings than conventional polymer composites filled with micro-scale fillers [1–4]. Graphene, particularly, have gained growing interest in the development of graphene-polymer-nanocomposites [5, 6]. Graphene consists of two-dimensional layered carbon sheet composed of sp² carbon atoms, which recently has gained much attention due to its unique combination of properties.

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Graphene is stronger than steel and conducts electricity, at room temperature, better than any other materials known by mankind [7]. Graphene sheets provide an alternative option to produce functional nanocomposites due to their excellent properties and the natural abundance of their precursor, graphite [8]. The oxidation of natural flake graphite can conduct to the formation of two-dimensional lamellae of graphene oxide (GO) with large number of randomly distributed hydrophilic functional groups, such as hydroxyl, carbonyl, epoxide, ether, ester, and carboxylate groups [9, 10]. These polar groups afforded GO with excellent possibilities to be compatibilized with other polar systems. The presence of oxygen functionalities in GO is very interesting from a chemical point of view in polymer nanocomposite materials since they provide reactive sites for chemical interactions with the matrix [11]. Due to the presence of oxygen functionalities, GO can easily disperse in organic solvents, water, and different matrixes. This is a major benefit when combining GO with polymer matrixes to enhance their mechanical and electrical properties to approach different advanced applications. Compared to layered nanoclays, GO has a lamellar structure with noticeable higher mechanical properties and higher surface area that maximize the interfacial contact between polymer and filler. Even GO is partially hydrophilic compared with graphene and can be readily dispersed in water, it is still difficult to disperse and exfoliate it into single layers in a solid polymer matrix. Extensive research has been done on the preparation and characterization of GO-polymer nanocomposites using different techniques to enhance the GO dispersion in the polymer matrix [12–16]. These techniques include solution processing, *in situ* polymerization, and melt blending techniques among others.

The major challenge has been the homogeneous dispersion of GO particles into the polymer matrix, which is influenced by the interfacial interactions between the GO nanosheets and the polymer-surrounding matrix. Several works related with chemical functionalization of graphene and GO reported an effective improvement in the dispersion of graphene into the polymer matrix [17–20]. In the literature, some papers report the preparation of isotactic polypropylene (iPP) nanocomposites with graphite or graphite oxide by different types of techniques: molten state blending, solid-state shear pulverization, and solution processes [21–23]. Due to the lack of polar groups into a polyolefin polymer, the dispersion and exfoliation of GO in this hydrophobic semicrystalline polyolefin matrix such, as PP and PE has become a very challenging task [24]. Non-polar polyolefins seem incapable of exfoliating and dispersing many types of fillers by themselves, and addition of an appropriate compatibilizer is required. The reaction of maleic anhydride-grafted PP (PPgMA) with an amine alcohol (DMAE) may be used to generate a PP grafted with amine-alcohol (PPgDMAE) that, in theory, may interact with the surface moieties of the GO nanostructures lowering their surface energy and enhancing

their dispersion on the polymer matrix. Amine-alcohol modified polyolefins have demonstrated their ability to exfoliate and disperse other types of fillers such as clay and silver nanoparticles [24–26]. The ability of PPgDMAE compatibilizer to exfoliate and disperse graphene nanoparticles and its effect on mechanical and electrical properties of PP/GO nanocomposite was examined in this work. The using of surfactants and functionalization to enhance dispersion of graphene particles in polymer matrix has been extensively investigated previously, but to the best of our knowledge, there are not many reported works related with the effect of using amine-alcohol grafted polyolefin as compatibilizer on the properties of PP/GO composite systems. In this work, we reported a strategy to improve dispersion and interfacial strength between filler and polymer matrix using a more effective functionalized polymer (amine-alcohol grafted PP).

EXPERIMENTAL

Materials

Natural graphite nanopowder flakes with particle size of 400 nm, 99.9% purity was purchased from Nanostructured and Amorphous Materials. Isotactic polypropylene homopolymer (Valtec PP-HP423, Melt flow index: 3.2 g/10 min) and maleic anhydride grafted polypropylene (PPgMA) (Polybond 3200; 1 wt% MA) were obtained from Indelpro, Mexico and Chemtura-Uniroyal Chemical Company, respectively. 2-[2-(dimethylamino)-ethoxy] ethanol (DMAE) and 2,6-di-*tert*-butyl-4-methylphenol (BHT, 99%) from Sigma-Aldrich, which were reagent grade and were used without further purification.

Preparation of Graphene Oxide

GO was synthesized from purified natural graphite by a modified Hummer's method [27]. In brief, 1 g of graphite nanopowder flakes was added to 100 mL of concentrated sulfuric acid and was subjected to sonication for 30 min using a Branson digital sonicator (S450D, 250 W, 30% amplitude) followed by the addition of 1 g of sodium nitrate (NaNO_3) in an ice bath. To this mixture, 6 g of potassium permanganate (KMnO_4) was slowly added under ice-cold conditions. The mixture was stirred for 2 h and the temperature was raised and maintained at 35°C in a water bath for another 0.5 h. Next, 46 mL of 70°C water was added drop-wise to the solution, and the temperature of the system was increased to 98°C. Finally, 140 mL of 70°C water was added, followed by 20 mL of a 30 wt% hydrogen peroxide (H_2O_2) solution to terminate the reaction. The synthesized graphite oxide was suspended in water and purified by dialysis to completely remove residual salts and acids. The resulting GO was dried overnight at 55°C under vacuum (40 mm Hg) to produce GO powder.

TABLE 1. Composite samples description

Sample	Method	PP (wt%)	PPgDMAE (wt%)	GO (wt%)
M-PP/PPgDMAE-15/GO-0.5	Melt-mixing	84.5	15	0.5
M-PP/PPgDMAE-15/GO-1.0	Melt-mixing	84	15	1.0
M-PP/PPgDMAE-15/GO-1.5	Melt-mixing	83.5	15	1.5
M-PP/PPgDMAE-25/GO-0.5	Melt-mixing	74.5	25	0.5
M-PP/PPgDMAE-25/GO-1.0	Melt-mixing	74	25	1.0
M-PP/PPgDMAE-25/GO-1.5	Melt-mixing	73.5	25	1.5
S-PP/PPgDMAE-15/GO-0.5	Solution-blending	84.5	15	0.5
S-PP/PPgDMAE-15/GO-1.0	Solution-blending	84	15	1.0
S-PP/PPgDMAE-15/GO-1.5	Solution-blending	83.5	15	1.5
S-PP/PPgDMAE-25/GO-0.5	Solution-blending	74.5	25	0.5
S-PP/PPgDMAE-25/GO-1.0	Solution-blending	74	25	1.0
S-PP/PPgDMAE-25/GO-1.5	Solution-blending	73.5	25	1.5

Preparation of Polypropylene-PPgDMAE-Graphene Oxide Nanocomposites

The preparation of the PPgDMAE compatibilizer was done using PPgMA previously dried at 90°C for 24 h and premixed in presence of BHT within a Banbury batch-mixing chamber (Brabender PL2000). After 2 min of mixing, the amino alcohol (DMAE) was added, in sealed PP bags, and the mixing was continued for 12 min at 170°C and 60 rpm, forming the PPgDMAE. Polypropylene (PP) composites based on graphene oxide and functionalized PP were prepared by two different methods: one consisted on direct melt blending, the three components with various amounts of GO (0.5, 1.0, and 1.5 wt%) using a Banbury batch mixing Chamber (Brabender PL2000). A temperature of 190°C, mixing time of 12 min and “cam” type rotors at a speed of 60 rpm were determined to be the more convenient processing conditions. In another method, GO and PPgDMAE were previously dissolved in Xylene at 110°C and subjected to sonication for 30 min using a Branson digital sonicator. The solvent was evaporated and the composite sample was dried and incorporated to PP by melt blending with various amounts of GO (0.5, 1.0, and 1.5 wt%) using a Banbury batch mixing Chamber. Melt blended samples were then pressed at 200°C for 2 min using a Carver press to prepare a 0.15-mm-thick sheet. The prepared nanocomposites are described in Table 1.

Characterization

The Fourier transform infrared spectra were recorded on a Nicolet 550 at 4 cm⁻¹ of resolution and 30 scans. X-ray diffraction was performed in a Siemens D5000 using Cu K α X-ray radiation. Raman spectra were obtained using a Horiba XploRA micro Raman spectrometer. The scanning electron microscopy (SEM) observations were carried out using a Top Con 510 SM. The surface wettability of graphite and GO samples was expressed in terms of the contact angles of water on the sample surface using a Contact Angle Meter Rame-hart-100-00 (NJ). Doubly distilled water was used in this analysis. Tensile properties were determined using an Instron model 4301 upgraded for computerized data acquisi-

tion. Thermogravimetric analyses (TGA) were performed in a TA Instruments TGAQ500 using a heating rate of 10°C/min, under a nitrogen atmosphere. The electrical properties of the polypropylene nanocomposite samples were determined using a two-probe method with a Meter Keithley 2400 (Keithley Instruments).

RESULTS AND DISCUSSION

Fourier Transform Infrared (FTIR) Characterization

Synthesis of graphene oxide (GO) from graphite using a modified Hummer's method, as described in the experimental section, produced a graphene material with very different oxygen-containing functional groups such as epoxy, phenolic, carbonyl, and carboxylic acid groups, among others, distributed through the basal plane and along the edges of the graphene sheets. Figure 1 show the

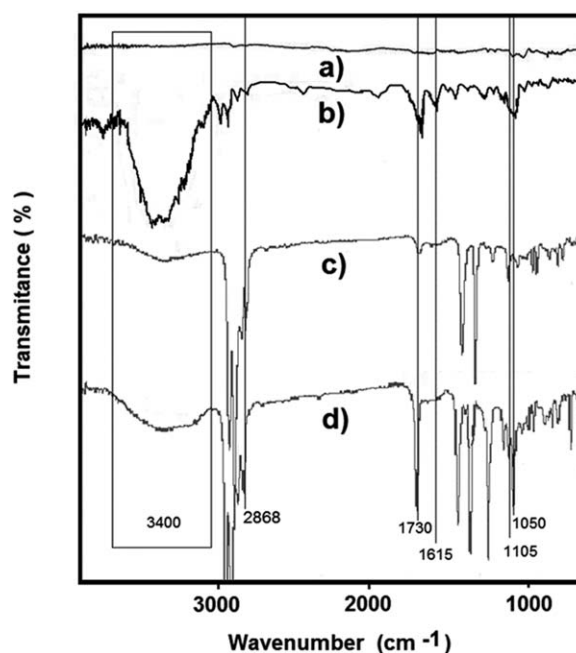


FIG. 1. FTIR of (a) graphite, (b) GO, (c) M-PP/PPgDMAE-15/GO-1.0, and (d) S-PP/PPgDMAE-15/GO-1.0.

Fourier transform infrared spectroscopic (FTIR) images for pristine graphite, GO, M-PP/PPgDMAE-15/GO-1.0, and S-PP/PPgDMAE-15/GO-1.0. It can be seen that while graphite sample (Fig. 1a) has not any significant stretching band, GO sample (Fig. 1b) has remarkable stretching bands associated with the oxygen-containing polar groups. In GO sample, a broad signal around $3,400\text{ cm}^{-1}$ associated with the hydroxyl groups stretching band as well with some water trapped between graphene layers can be observed. The abortion band located near $1,730\text{ cm}^{-1}$ is associated with the ketone carbonyl groups and the vibrational double bond ($\text{C}=\text{C}$) of graphite sheets appears around $1,615\text{ cm}^{-1}$. The signal around $1,050\text{ cm}^{-1}$ is associated with the epoxide groups stretching band. However, PP/GO samples show the characteristic bands of PP and GO but sample prepared by solution blending (Fig. 1d) shows more pronounced absorption bands associated with the oxygen-containing groups from the GO compared with the sample prepared by melt mixing (Fig. 1c). This suggest that GO has been reduced after melt mixing in such high processing temperatures of PP composite. The GO reduction by melt mixing are responsible of the less notorious oxygen absorption bands at $3,400\text{ cm}^{-1}$ compared with the sample prepared by solution procedure. These samples also show peaks around $2,868$ and $1,105\text{ cm}^{-1}$ that can be related with the C-H , C-O , and N-H bonds that could suggest possible interactions between PPgDMAE and GO.

Raman Characterization

Raman spectroscopy is a commonly used tool for the analysis of graphite-like materials such as carbon nanotubes (CNTs) and graphene. Conjugated and carbon-carbon double bonds can result in high Raman intensities. Representative Raman spectra of pristine graphite, GO, and nanocomposite of GO and PP prepared by both methods (M-PP/PPgDMAE-15/GO-1.0 and S-PP/PPgDMAE-15/GO-1.0) are shown in Fig. 2. The D peak ($1,340\text{--}1,350\text{ cm}^{-1}$) in the Raman spectra corresponds to a disorder-induced Raman mode and is associated with the breathing modes in C-C ring structures. The D band indicates the presence of disordered graphite and its intensity can indicate the amount of disorder in graphene. The G mode ($1,560\text{--}1,570\text{ cm}^{-1}$) is assigned to C-C (sp^2) bond stretching vibrations [28]. It can be seen that the D band appears at the same position for all the samples but this signal is broader for GO and nanocomposite samples compared with graphite sample. The Raman spectrum of GO and nanocomposite exhibits a D band at $1,344\text{ cm}^{-1}$, corresponding to sp^3 -hybridized carbons. The intensity of the D band is almost the same as that of G band for GO samples and slightly higher in GO/PP composite, revealing the presence of certain defects in the as-prepared samples. GO/PP nanocomposite sample shows a slightly higher intensity D band ($1,344\text{ cm}^{-1}$) than G band at $1,578\text{ cm}^{-1}$ compared with GO sample. The intensity

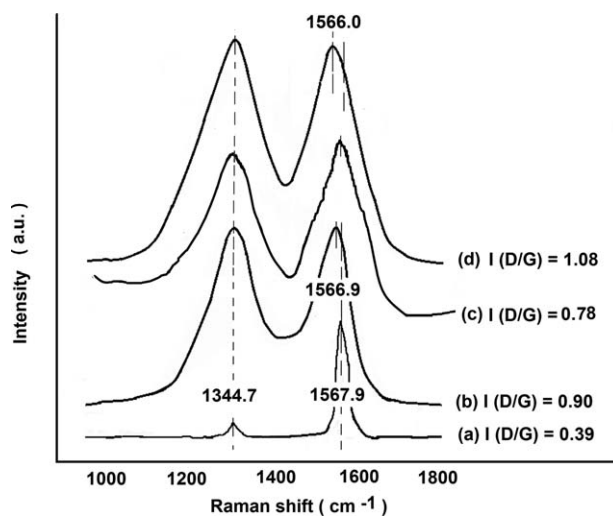


FIG. 2. Raman spectra of (a) graphite, (b) GO, (c) M-PP/PPgDMAE-15/GO-1.0, and (d) S-PP/PPgDMAE-15/GO-1.0.

ratio of D and G bands ($I_{\text{D/G}}$) has been associated with the disordered and ordered crystal structures of carbon [29, 30]. This ratio was calculated from the D peak and G peak at the greatest height in the spectra shown in Fig. 2. It can be seen that this ratio is increased from 0.39 for pristine graphite to 0.90 for GO, 0.78 for M-PP/PPgDMAE-15/GO-1.0, and 1.08 for S-PP/PPgDMAE-15/GO-1.0 composite. It has been reported [28, 29] that a large increase in this ratio is caused by covalent functionalization of the functional groups within the internal domain of graphene platelets. The large increase in this ratio also is associated with an excess of defects in the graphitic lattice and a decrease in the average size of the in-plane sp^2 domain. These defects could be associated with the interactions between GO and amine groups of compatibilizer and also with the partial thermal reduction during solution blending [30, 31]. It also can be seen that M-PP/PPgDMAE-15/GO-1.0 sample shows a lower increase in this ratio compared with the solution samples this indicates a more pronounced reduction of GO during melt mixing at high PP processing temperatures. It can be appreciated a slight downshifted of G band for the solution sample compared with GO and graphite samples. It has been reported [32, 33] that the G band of graphene is downshifted when a stretching is applied because of the elongation of the carbon-carbon bonds. This elongation can be associated with the interactions between GO and compatibilizer.

XRD Characterization

The XRD patterns of PP, graphite, GO, and PP/GO composites are plotted in Fig. 3. X-ray diffraction results of PP sample (Fig. 3a) revealed the formation of characteristic PP α crystalline phases that are corroborated by the appearance of the peaks at $2\theta = 14.0^\circ$ (110); 17.3° (040); 18.7° (130); and 22.0° (111). No β -form crystalline

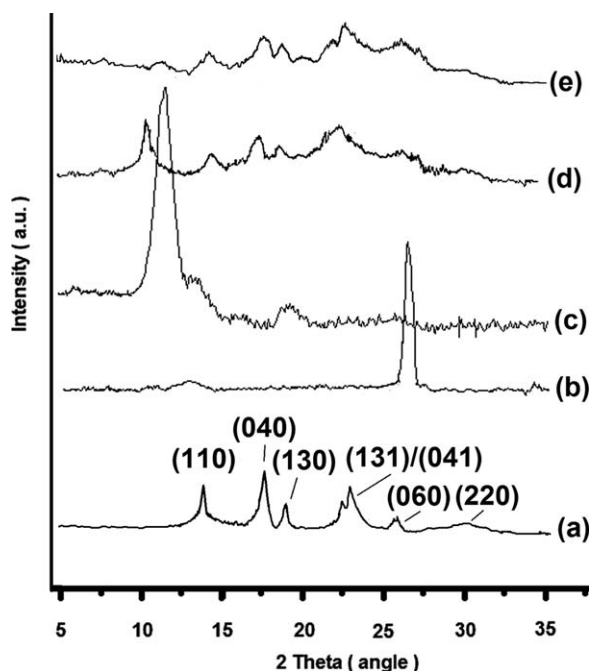


FIG. 3. The X-ray diffraction spectra of (a) neat PP, (b) graphite, (c) GO, (d) S-PP/PPgDMAE-15/GO-1.0, and (e) M-PP/PPgDMAE-15/GO-1.0.

phase was found in the PP/GO composites indicating that GO may be just acting as an α -nucleating agent for PP crystallization. A sharp peak near 26° observed in the graphite sample (Fig. 3b) is shifted to lower angles (near 12°) for the GO sample (Fig. 3c). These sharp peaks can be related with the high degree of order and the shifting to lower angles is indicative of higher interlayer spacing. This peak shifting to 12° in GO sample corresponds to an

interlayer spacing of about 0.73 nm, calculated via the Bragg equation, compared to 0.35 nm obtained for pristine graphite. The severely weakened intensity of this peak and its shifting to 10° (0.88 nm) for the PP/GO nanocomposite solution sample, S-PP/PPgDMAE-15/GO-1.0, (Fig. 3d) indicates a disorder in the remaining stacks caused by the interactions between PPgDMAE and GO sheets. The melt mixing sample XRD pattern (Fig. 3e), besides the PP characteristic diffraction peaks, do not show any notorious diffraction peak, only a slight shoulder around 20° indicating a reduced interlayer spacing, probably due to the elimination of most oxygen-containing groups during the partial reduction at high melting temperatures.

SEM and TEM Characterization

The morphology of the GO and PP/GO nanocomposites prepared by both melt and solution blending method, characterized by scanning electron microscopy (SEM) and transmission electron microscopy (TEM) is shown in Figs. 4 and 5. The morphology of GO samples (Fig. 4a) shows multilayer GO platelets. The fractured PP sample (Fig. 4b) shows the characteristic smooth brittle fracture morphology while the GO composites show a very different morphology with noticeable GO particles of more than $10\ \mu\text{m}$ for the melt mixing prepared sample (Fig. 4c) and $5\ \mu\text{m}$ for the solution sample. It also can be seen more GO particle aggregation in the met mixing samples while in the solution sample (Fig. 4d) a stretched polymer coating around the GO particles can be seen. This less particle aggregation observed in solution sample can be attributed to the better dispersion induced for the polymer

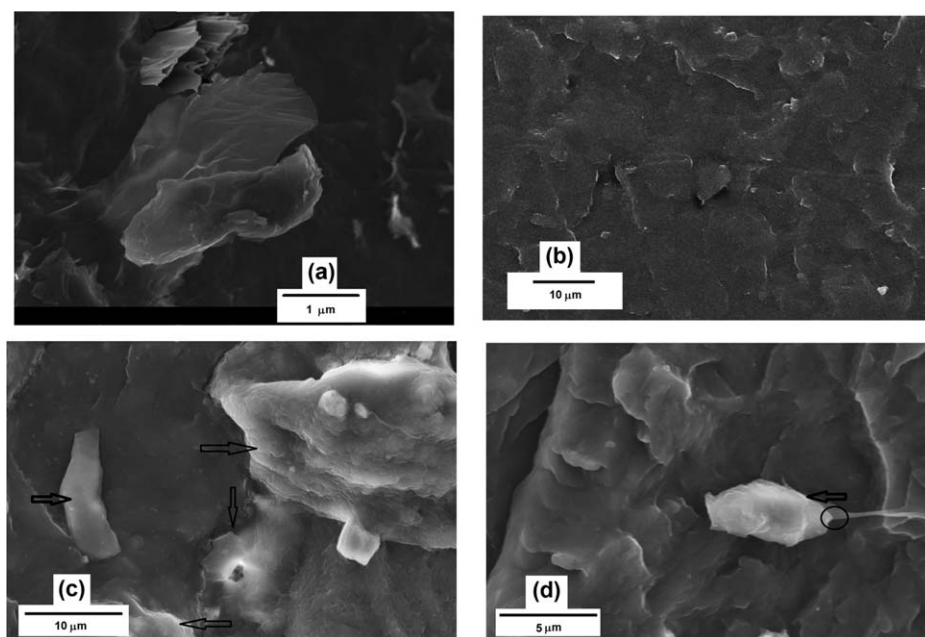


FIG. 4. SEM images of the cryogenically fractured surface of (a) GO, (b) neat PP, (c) M-PP/PPgDMAE-15/GO-1.0, and (d) S-PP/PPgDMAE-15/GO-1.0.

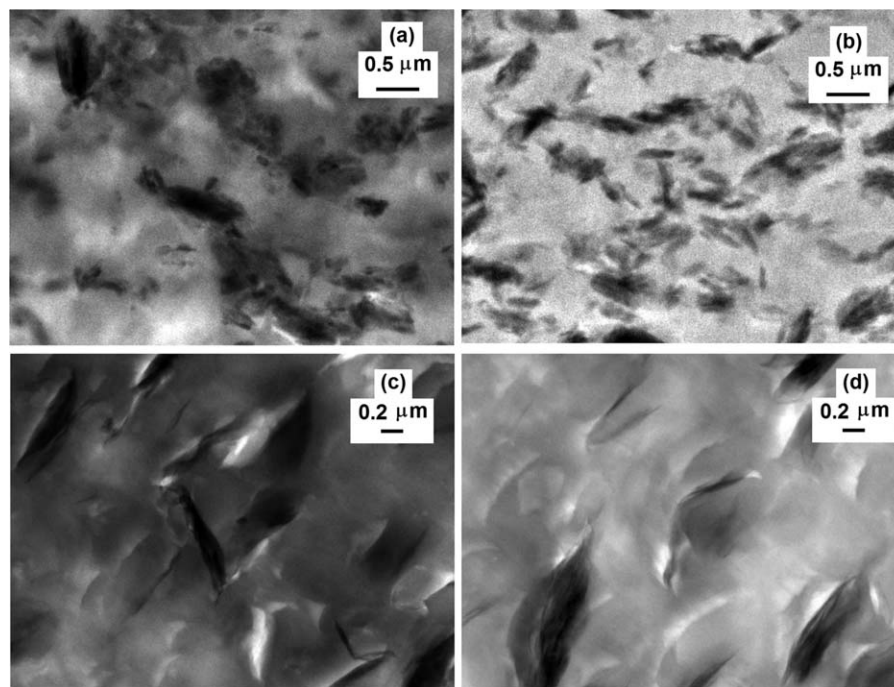


FIG. 5. TEM images for (a) and (c) M-PP/PPgDMAE-15/GO-1.0 and (b) and (d) S-PP/PPgDMAE-15/GO-1.0.

compatibilizer in the solution method. This was confirmed by TEM images from Fig. 5. Figure 5a shows noticeable GO agglomerated particles for the melt-mixing sample, while the solution-blending sample (Fig. 5b) shows less particle aggregates with a more homogeneous GO particle dispersion. At higher magnifications, the difference between both methods are more perceptible. Since the melt mixing sample (Fig. 5c) shows the GO particles in an agglomerated morphology, the solution blending sample, on the contrary shows a more exfoliated GO particles into the polymer matrix (Fig. 5d). This could be an indicative of the enhancement in compatibility between GO and PP matrix when using the solution blending method. Several authors have [34, 35] obtained similar results with PP/GO composites compatibilized with different types of compatibilizers such as PP-g-MA (polypropylene grafted maleic anhydride) and PP-g-GOs (polypropylene grafted GO sheets).

Mechanical Characterization

The homogeneous dispersion of GO nanosheets in the matrix is very crucial to improve the mechanical property of the GO nanocomposites [32, 36]. The tensile strength and tensile modulus of PP/GO nanocomposite samples prepared by solution and melt mixing methods are shown in Fig. 6. The incorporation of GO increases the tensile strength and its enhanced when the compatibilizer is changed from 15 to 25 wt%. In the melt-mixing method, a plateau and even a slight reduction on this property was observed when the GO content was increased. In the solution-blending method, higher values on tensile

strength with the increase on GO content was observed specially for higher compatibilizer contents. The tensile modulus in samples prepared by melt mixing is increased only at lower GO content (0.5 wt%) but it shows a noticeable reduction for higher GO contents for both compatibilizer contents. This reduction in modulus as well as in tensile strength at higher GO contents using melt mixing method indicates that the shear rate obtained with the internal mixer is not enough to promote a fair

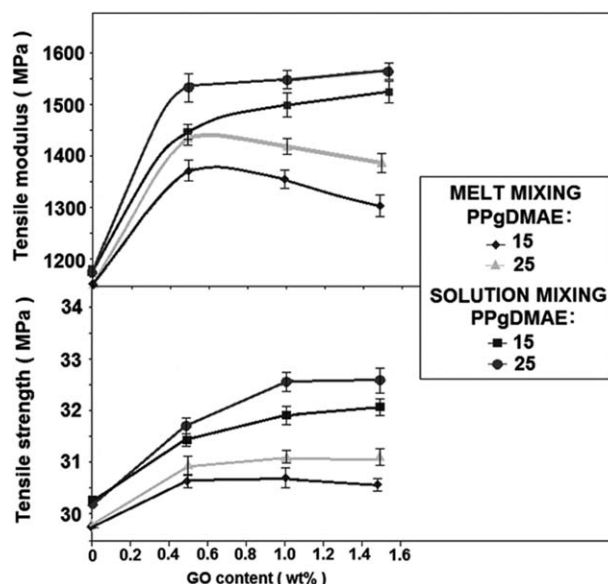


FIG. 6. Variation of tensile strength and modulus with the GO content of PP/GO nanocomposites prepared by solution blending and direct melt mixing.

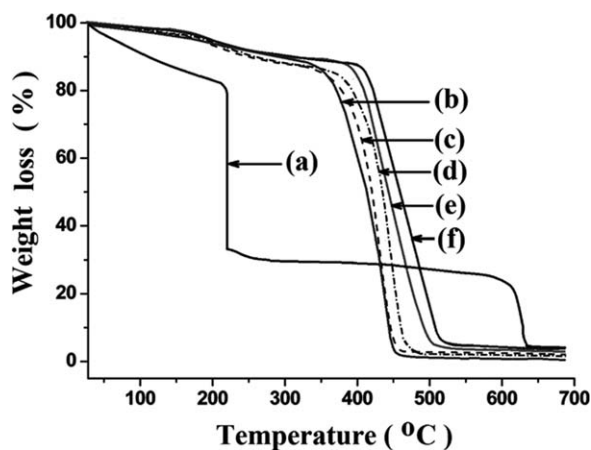


FIG. 7. TGA curves of (a) GO, (b) PP, (c) M-PP/PPgDMAE-15/GO-1.0, (d) M-PP/PPgDMAE-25/GO-1.0, (e) S-PP/PPgDMAE-15/GO-1.0 and (f) S-PP/PPgDMAE-25/GO-1.0.

interaction between GO and compatibilizer. This indicates that the shear stress obtained by batch mixer are not enough to induce a good particle dispersion compared with other mixing methods such as twin screw extruders. However, samples prepared by solution blending show a notorious increase in modulus when the GO content is increased, especially with higher compatibilizer contents. It can be seen that at higher GO contents the difference in this property between the two compatibilizer contents is minimal indicating that 15 wt% of compatibilizer is enough to obtain an acceptable value of tensile modulus. The improved mechanical properties obtained by solution blending could be attributed to a more homogeneous dispersion of GO and the large interfacial area that this method can promote, enhancing the strong interaction between GO and PP matrix when using PPgDMAE as compatibilizer.

TGA Characterization

The homogeneous dispersion of GO nanoparticles is also an important factor to enhance the thermal stability of PP/GO nanocomposites. Figure 7 shows the degradation behavior determined by TGA of GO and PP/GO nanocomposite samples, prepared by solution and melt mixing methods. The initial decomposition temperature ($T_{initial}$), defined as the temperature at 5 wt% weight loss and the maximum temperature (T_{max}) defined as the temperature at maximum rate of weight loss. It can be seen that for GO sample (Fig. 7a), the weight loss starts from 150 to 225°C indicating the abundant oxidative groups in this sample. PP/GO composite samples show more stable behavior than PP (Fig. 7b) suggesting that the incorporation of GO to PP matrix can greatly enhance the thermal stability. It is observed that the solution mixing has a more evident increase in thermal stability compared with melt mixing. It also can be observed that an increase in the compatibilizer agent content enhance the

degradation temperature of the composite. The T_{max} of the composites, compared with the corresponding value of PP, are increased by 15 and 20°C for 15 and 25 wt% of compatibilizer in samples prepared by melt mixing and by 30 and 40°C for 15 and 25 wt% of compatibilizer in samples prepared by solution blending. This increase in thermal stability of PP/GO samples is probably attributed to the GO that may be acting as physical barrier that mitigates the escape of decomposed volatiles and degrading products during pyrolysis enhancing the carbonization of the polymer matrix. The high thermal conductivity of graphene also may be acting as an efficient heat dissipater that dissipates external heat more uniformly and efficiently avoiding the accumulation of heat and reducing the thermal degradation of the sample [37]. It can be concluded that the even dispersion of GO in PP/GO samples achieved by the incorporation of PPgDMAE compatibilizer and prepared by solution blending enhances the thermal stability more than the melt mixing method. This could be reasonably explained by enhanced interfacial interaction between graphene nanosheets and PP matrix induced by compatibilizer and mixing method.

Electrical Characterization

Finally, the effect of compatibilizer and preparation method on the electrical properties of PP/GO nanocomposites was investigated. The obtained results are summarized in Fig. 8. As expected, when increasing the GO content a drastic decrease in electrical volume resistivity values was observed. Perhaps the high processing temperature of GO during the preparation of PP nanocomposite samples in the internal mixer (190°C) resulted in the removal of oxygen functionalities and thereby causing the observed low electrical resistivity values in PP/GO nanocomposites. This can be explained with the better GO dispersion that is obtained with the solution blending that enhances the interfacial interactions between GO and compatibilizer, which in turn, may result in the lowering of electrical percolation threshold and lower electrical resistivity. This effect is more pronounced when higher

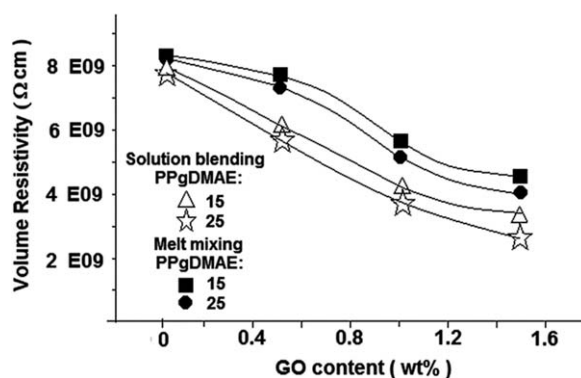


FIG. 8. Variation in electrical resistivity of PP/GO nanocomposites prepared by solution blending and direct melt mixing.

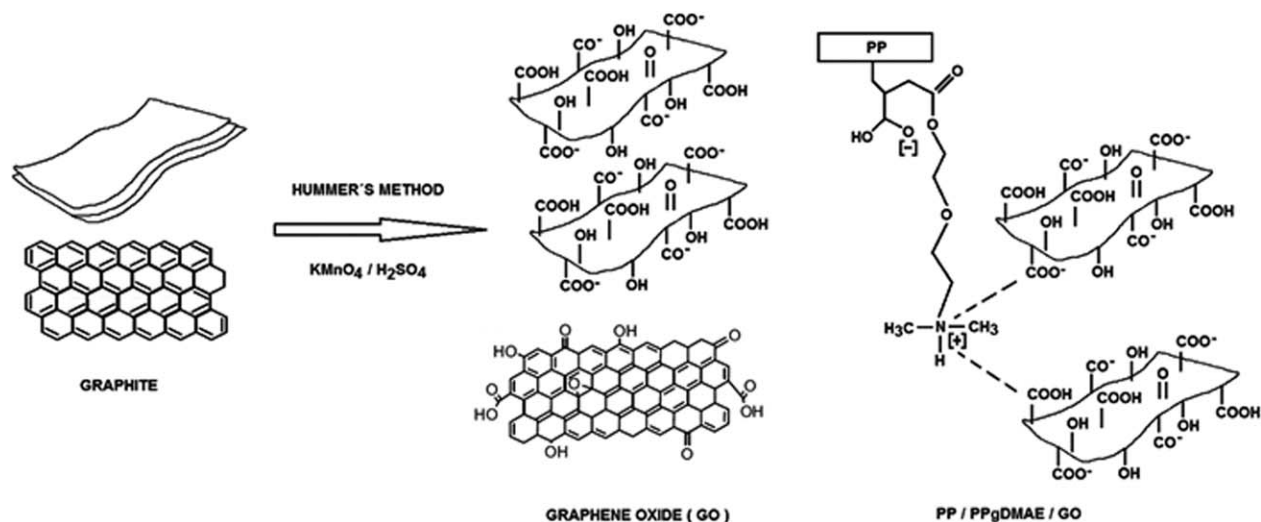


FIG. 9. Possible interaction mechanism between PP matrix and GO by using PPgDMAE as compatibilizer.

compatibilizer contents are used. This can be explained with the better GO dispersion that is obtained with the solution blending that enhances the interfacial interactions between GO and compatibilizer, which in turn, may result in the lowering of electrical percolation threshold and lower electrical resistivity. Li et al., [38] have reported similar electrical resistivity behavior with graphene nanoplatelets. The scheme that describes the interactions between GO and PP by using PPgDMAE as compatibilizer is summarized in Fig. 9. In a first step, the graphite was oxidized by Hummer's method to produce GO. Then, the amine alcohol modified PP compatibilizer were assembled with GO forming strong interactions between polar oxidized moieties in GO with the amine polar group on the compatibilizer. GO contains numerous oxygenous polar groups both on the basal planes as well as on the edges of graphene sheets [9, 10], thus it is logical to assume good compatibility between GO and compatibilizer. It is reported [39] that the epoxide groups in GO can react with amine groups via a ring-opening reaction. As well as the amido linkage between amine groups and carboxylic acid on GO that may be formed during melt mixing [40]. These strong interfacial interactions could be responsible for the enhancement in mechanical, thermal, and electrical properties observed in this work.

CONCLUSIONS

In this study, the influence of incorporating PPgDMAE into PP/GO nanocomposites and the preparation method was investigated. Successful oxidation of graphite to obtain graphene oxide (GO) was confirmed by FTIR and contact angle measurements. XRD, FTIR, and Raman spectroscopy revealed the interactions between GO and PPgDMAE compatibilizer. Polypropylene/GO nanocomposites using PPgDMAE with homogeneous dispersion of GO nanosheets and enhanced nanofiller-matrix interfacial

interactions were prepared through solution blending and direct melt mixing. Significant improvements in mechanical, thermal, and electrical properties were observed when PPgDMAE was incorporated and when the solution blending was used due to strong interfacial interactions between the polymer matrix and GO when using PPgDMAE. Using of this type of compatibilizer can offer a better option for obtaining PP/GO nanocomposites for many practical applications.

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CONFLICTS OF INTEREST

The authors declare that there is no conflict of interest.

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